BIOCHEMISTRY CUMULATIVE EXAM

December 5, 1992

Electron transfer processes are essential to many biochemical processes including respiration and photosynthesis. The 1992 Nobel Prize in Chemistry was awarded to an individual who is responsible for the development of theoretical analysis of electron transfer processes. Although the theoretical framework was laid down over 30 years ago, this theory is widely used today to analyze many types of electron transfer including biological systems. A generalized equation summarizing the essence of this theory is shown below.

$$k_{el} = 10^{13} \exp[-\beta(r)] \exp[-\Delta G^*/RT] \ s^{-1}$$

where

$$\Delta G^* = (\lambda + \Delta G^0)^2/4\lambda$$

k_{el} = first order rate constant for electron transfer

λ = reorganization energy

β = scalar relating rate to distance (r), typically 1.0 A^{-1}

(i) [10 points] Who was the recipient of the 1992 Nobel Prize in Chemistry? At what institution is he presently located?

(ii) [20 points] Sketch the dependence of the rate constant for electron transfer on the driving force, ΔG^0, at constant r. (Hint: it may help to take logarithms of the above equation.) We often tell freshman chemistry students that there is no relationship between reaction rate and thermodynamic parameters such as ΔG^0. Why do electron transfer reactions appear to violate this fundamental statement?

(iii) [20 points] At fixed distance (r = constant), what mathematical relationship between ΔG^0 and λ would yield the maximum electron transfer rate? Show where λ is on your sketch for part (i).

(iv) [15 points] Sketch the dependence of the rate constant for electron transfer on distance between the redox centers at constant driving force. Again, taking logarithms of the above equation might be useful.

(v) [15 points] What is the relationship between λ and ΔG^*, the free energy of activation, when in the absence of driving force, i.e., when ΔG^0, is equal to zero.

(vi) [20 points] A given redox reaction has a ΔG^0 of -0.10 eV and a λ of 1.0 eV. The measured first order rate constant at 298 K is 10,000 s^{-1}. Use these values to estimate the distance at which electron transfer occurs in this system. ($R = 8.59 \times 10^{-5}$ eV/mole-K).